

SIMATS ENGINEERING
ASSESSMENT TOOL 1: INDUSTRY PROBLEM SOLVING TASKS

COURSE NAME:	Programming in Java	COURSE CODE:	CSA0915- SLOT B
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Assessment Tasks

2. Industry Scenario:

A bank wants to develop an Online Fund Transfer Interface for its customer service department.

Question:

Design and develop a Java AWT application that allows a customer to transfer money from one account to another. The interface should contain fields for Account Number, Beneficiary Account Number, Account Holder Name, Transfer Amount, and Transaction Type.

Use appropriate AWT components such as Label, TextField, Choice, Button, TextArea, and Frame. When the Transfer button is clicked, validate the input, check whether the transfer amount is valid, and display a transaction summary with the transferred amount and transaction status. Provide Clear and Exit buttons.

Requirements:

1. Design the complete AWT interface.
2. Select suitable containers and components.
3. Implement event handling.
4. Validate account details and transfer amount.
5. Display successful/failed transaction status.
6. Implement Clear and Exit operations.

Test Cases:

tc1_input- Account=1001, Beneficiary=2001, Amount=5000
tc1_output- Transfer Successful
tc2_input—Account=1002, Beneficiary=2002, Amount=10000
tc2_output- Transfer Successful
tc3_input- Transfer Amount = 0
tc3_output- Invalid Transfer Amount
tc4_input- Beneficiary Account empty
tc4_output- Beneficiary Account Required
tc5_input-Account Number empty
tc5_output-Account Number Required

CODE:

```
import java.awt.*;
```

```
import java.awt.event.*;
```

```
public class FundTransfer extends Frame implements ActionListener {
```

```
    Label accountLabel, beneficiaryLabel, nameLabel, amountLabel, typeLabel;
```

```
    TextField accountField, beneficiaryField, nameField, amountField;
```

```
    Choice typeChoice;
```

```
    Button transferButton, clearButton, exitButton;
```

```
    TextArea summaryArea;
```

```
    public FundTransfer() {
```

```
        setTitle("Online Fund Transfer Interface");
```

```
        setSize(600, 500);
```

```
        setLayout(new BorderLayout(10, 10));
```

```
        Label title = new Label("ONLINE FUND TRANSFER", Label.CENTER);
```

```
        title.setFont(new Font("Arial", Font.BOLD, 20));
```

```
        add(title, BorderLayout.NORTH);
```

```
        Panel formPanel = new Panel();
```

```
        formPanel.setLayout(new GridLayout(5, 2, 10, 10));
```

```
        accountLabel = new Label("Account Number:");
```

```
        accountField = new TextField();
```

```
beneficiaryLabel = new Label("Beneficiary Account:");
```

```
beneficiaryField = new TextField();
```

```
nameLabel = new Label("Account Holder Name:");
```

```
nameField = new TextField();
```

```
amountLabel = new Label("Transfer Amount:");
```

```
amountField = new TextField();
```

```
typeLabel = new Label("Transaction Type:");
```

```
typeChoice = new Choice();
```

```
typeChoice.add("Select Type");
```

```
typeChoice.add("NEFT");
```

```
typeChoice.add("RTGS");
```

```
typeChoice.add("IMPS");
```

```
formPanel.add(accountLabel);
```

```
formPanel.add(accountField);
```

```
formPanel.add(beneficiaryLabel);
```

```
formPanel.add(beneficiaryField);
```

```
formPanel.add(nameLabel);
```

```
formPanel.add(nameField);
```

```
formPanel.add(amountLabel);
```

```
formPanel.add(amountField);
```

```
formPanel.add(typeLabel);
```

```
formPanel.add(typeChoice);
```

```
add(formPanel, BorderLayout.CENTER);
```

```
Panel bottomPanel = new Panel(new BorderLayout());
```

```
Panel buttonPanel = new Panel();
```

```
transferButton = new Button("Transfer");
```

```
clearButton = new Button("Clear");
```

```
exitButton = new Button("Exit");
```

```
transferButton.addActionListener(this);
```

```
clearButton.addActionListener(this);
```

```
exitButton.addActionListener(this);
```

```
buttonPanel.add(transferButton);
```

```
buttonPanel.add(clearButton);
```

```
buttonPanel.add(exitButton);
```

```
bottomPanel.add(buttonPanel, BorderLayout.NORTH);
```

```
summaryArea = new TextArea("", 10, 60);

summaryArea.setEditable(false);

bottomPanel.add(summaryArea, BorderLayout.CENTER);

add(bottomPanel, BorderLayout.SOUTH);

addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent e) {
        System.exit(0);
    }
});

setVisible(true);
}

public void actionPerformed(ActionEvent e) {

    if (e.getSource() == transferButton) {
        transferMoney();
    }

    if (e.getSource() == clearButton) {
        clearFields();
    }
}
```

```
if (e.getSource() == exitButton) {  
    System.exit(0);  
}  
}
```

```
private void transferMoney() {
```

```
    String account = accountField.getText().trim();  
    String beneficiary = beneficiaryField.getText().trim();  
    String name = nameField.getText().trim();  
    String amountText = amountField.getText().trim();
```

```
    if (account.isEmpty()) {  
        summaryArea.setText("Account Number Required");  
        return;  
    }
```

```
    if (beneficiary.isEmpty()) {  
        summaryArea.setText("Beneficiary Account Required");  
        return;  
    }
```

```
    if (name.isEmpty()) {  
        summaryArea.setText("Account Holder Name Required");  
        return;  
    }
```

```
if (amountText.isEmpty()) {  
    summaryArea.setText("Transfer Amount Required");  
    return;  
}
```

```
double amount;
```

```
try {  
    amount = Double.parseDouble(amountText);  
  
    if (amount <= 0) {  
        summaryArea.setText("Invalid Transfer Amount");  
        return;  
    }  
}
```

```
} catch (NumberFormatException ex) {  
    summaryArea.setText("Invalid Transfer Amount");  
    return;  
}
```

```
if (typeChoice.getSelectedIndex() == 0) {  
    summaryArea.setText("Transaction Type Required");  
    return;  
}
```

```
String transactionType = typeChoice.getSelectedItemId();

summaryArea.setText(

    "Transfer Successful\n\n" +

    "Account Number    : " + account + "\n" +

    "Beneficiary Account : " + beneficiary + "\n" +

    "Account Holder Name : " + name + "\n" +

    "Transfer Amount    : Rs. " + amount + "\n" +

    "Transaction Type   : " + transactionType + "\n" +

    "Transaction Status : Successful"

);

}
```

```
private void clearFields() {

    accountField.setText("");
    beneficiaryField.setText("");
    nameField.setText("");
    amountField.setText("");
    typeChoice.select(0);
    summaryArea.setText("");

}
```

```
public static void main(String[] args) {

    new FundTransfer();

}
```


}